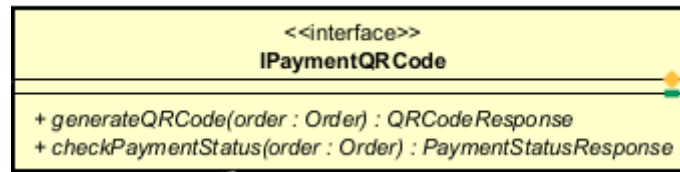


Design for interface “IPaymentQRCode”



Description: IPaymentQRCode defines the standard operations for QR code-based payment processing. It provides methods to generate QR codes and check payment status for orders. Concrete payment subsystems (in our case it is VietQR) must implement this interface.

Table 2. Operation design

#	Name	Return type	Description (purpose)
1	generateQRCode	QRCodeResponse	Generate a QR code for the given order and return QR payment information.
2	checkPaymentStatus	PaymentStatusResponse	Check the payment status of the given order and return the payment result.

Parameter:

generateQRCode(order : Order) : QRCodeResponse

- order: order object used to generate the payment QR code based on order information such as order ID and total amount.

checkPaymentStatus(order : Order) : PaymentStatusResponse

- order: order object used to check the payment status of the corresponding transaction.

Method:

generateQRCode(order : Order)

- Defines the method for generating a payment QR code based on the provided order.
- Must be implemented by the concrete payment QR controller class.
- Returns the QR payment information for display to the customer.

checkPaymentStatus(order : Order)

- Defines the method for checking the payment status of the provided order.
- Must be implemented by the concrete payment QR controller class.
- Returns the current payment status information.

Design for class “VietQRController”

VietQRController	
- boundary : VietQRBoundary - accessToken : String - tokenExpiryTime : long - vietqrUsername : String - vietqrPassword : String - bankCode : String - accountNo : String - accountName : String	
+ generateQRCode(order : Order) : QRCode + checkPaymentStatus(order : Order) : QRPaymentStatus - getValidAccessToken() : String	

Table 1. Attribute design

#	Name	Data type	Default value	Description
1	boundary	VietQRBoundary		Boundary object used to send HTTP requests to VietQR APIs.
2	accessToken	String		Current VietQR access token used for authenticated API calls.

3	tokenExpiryTime	long		Expiration time of the current access token in milliseconds.
4	vietqrUsername	String		Username of VietQR sandbox
5	vietqrPassword	String		Password of VietQR sandbox
6	bankCode	String		Bank code of the receiver account used to generate QR payment.
7	accountNo	String		Bank account number used to receive customer payment.
8	accountName	String		Bank account owner name

Table 2. Operation design

#	Name	Return type	Description (purpose)
---	------	-------------	-----------------------

1	generateQRCode	QRCodeResponse	Generate a QR code for the given order and return the QR payment information.
2	checkPaymentStatus	PaymentStatusResponse	Check the payment status of the given order and return the result.
3	getValidAccessToken	String	Return a valid VietQR access token, reusing the current one if it is still valid or requesting a new one if needed.

Parameter:

- generateQRCode(order : Order) : QRCode
 - order: the order used to generate the payment QR code
- checkPaymentStatus(order : Order) : PaymentStatusResponse
 - order: the order used to check the payment status

Exception

- InvalidTokenException: if VietQR returns an invalid access token.
- PaymentException: if QR generation or payment status checking cannot be completed.
- UnknownException: if an unexpected error occurs during VietQR API communication or response handling.

Design for class “QRCode”

QRCode
- bankName : String - bankCode : String - qrCode : String - qrLink : String - bankAccount : String
+ parseQRCodeResponse(response : String) : void + getQrCode() : String + getQrLink() : String + getBankCode() : String + getBankName() : String + getBankAccount() : String

Table 1. Attribute design

#	Name	Data type	Default value	Description
---	------	-----------	---------------	-------------

1	qrCode	String		QR code string returned from VietQR API, used for displaying or generating the QR image.
2	qrLink	String		Link to view the generated QR code.
3	bankCode	String		Code of the receiving bank.
4	bankName	String		Name of the receiving bank.
5	bankAccount	String		Bank account number used for payment.

Design for class “QRGenerateRequest”

QRGenerateRequest	
- bankCode : String - bankAccount : String - userBankName : String - content : String - qrType : int - amount : long - orderId : String - transType : String	
~ QRGenerateRequest(bankCode : String, bankAccount : String, userBankName : String, content : String, amount : long, orderId : String) ~ buildRequestString() : String	

Table 1. Attribute design

#	Name	Data type	Default value	Description
1	bankCode	String		Code of the receiving bank used for VietQR payment generation.
2	bankAccount	String		Bank account number used to receive payment.

3	userBankName	String		Name of the bank account owner.
4	content	String		Payment transfer content / transaction description.
5	qrType	int		Type of QR code (dynamic, static, or semi-dynamic).
6	amount	long		Payment amount of the order.
7	orderId	String		ID of the order associated with the payment.
8	transType	String		Type of transaction (e.g. credit transaction).

Table 2. Operation design

#	Name	Return type	Description (purpose)
1	QRGenerateRequest	constructor	Initialize request object with payment

			information for VietQR API.
2	buildRequestString	String	Convert request attributes into request string / JSON format for VietQR API call.

Parameter:

- QRGenerateRequest(bankCode : String, bankAccount : String, userBankName : String, content : String, amount : long, orderId : String)
 - bankCode: code of the receiving bank
 - bankAccount: bank account number for payment
 - userBankName: account owner name
 - content: payment transfer content
 - amount: payment amount
 - orderId: ID of the corresponding order

Method:

buildRequestString() : String

- Converts all stored attribute values into the request format required by the VietQR API.
- Constructs a formatted JSON / request string containing all necessary payment and bank information.
- Returns the final request string to be sent by the boundary layer.

Design for class “QRAccessTokenResponse”

QRAccessTokenResponse
- accessToken : String - tokenType : String - expiresIn : int
~ parseResponseString(response : String) : void ~ getAccessToken() : String ~ getExpiresIn() : int ~ isValid() : boolean

Table 1. Attribute design

#	Name	Data type	Default value	Description
1	accessToken	String		Access token returned from VietQR API, used for authenticated requests.
2	tokenType	String		Type of token returned by VietQR (e.g. Bearer).
3	expiresIn	int		Token expiration time returned by VietQR API, usually in seconds.

Table 2. Operation design

#	Name	Return type	Description (purpose)
1	parseResponseString	void	Parse raw VietQR token response string and extract token information.
2	getAccessToken	String	Return the parsed access token.
3	getExpiresIn	int	Return the token expiration time.
4	isValid	boolean	Check whether the parsed token response is valid.

Parameter:

- parseResponseString(response : String) : void
- response: response string returned from VietQR token API

Design for class “QRAccessTokenRequest”

QRAccessTokenRequest
- username : String - password : String
~ QRAccessTokenRequest(username : String, password : String) ~ buildAuthorizationHeader() : String

Table 1. Attribute design

#	Name	Data type	Default value	Description
1	username	String		VietQR account username used to request access token.
2	password	String		VietQR account password used to request access token.

Table 2. Operation design

#	Name	Return type	Description (purpose)
1	QRAccessTokenRequest	constructor	Initialize token request object with username and password.
2	buildAuthorizationHeader	String	Build authorization header string for VietQR token API request.

Parameter:

- QRAccessTokenRequest(username : String, password : String)
 - username: VietQR account username
 - password: VietQR account password

Method:

buildAuthorizationHeader() : String

- Generates the authorization header required for authenticating with the VietQR API.
- Encodes the username and password into the appropriate format (e.g., Base64 for Basic Authentication).
- Constructs the final authorization string

- Returns the authorization header string to be used in API requests sent by the boundary layer.

Design for class “VietQRBoundary”

VietQRBoundary	
- <u>GET_TOKEN_URL</u> : String - <u>GENERATE_QR_URL</u> : String - <u>TEST_CALLBACK_URL</u> : String - httpClient : HttpClient	
~ getAccessToken(authorizationHeader : String) : String ~ generateQRCode(accessToken : String, requestString : String) : String ~ checkPaymentStatus(accessToken : String, requestString : String) : String	

Table 1. Attribute design

#	Name	Data type	Default value	Description
1	GET_TOKEN_URL	String		URL of the VietQR API endpoint used to request an access token.

2	GENERATE_QR_URL	String		URL of the VietQR API endpoint used to generate a payment QR code.
3	TEST_CALLBACK_URL	String		URL of the VietQR API endpoint used to simulate or check payment callback status.
4	httpClient	HttpClient		HTTP client used to send requests to VietQR APIs.

Table 2. Operation design

#	Name	Return type	Description (purpose)
1	getAccessToken	String	Send a request to VietQR to obtain an access token and return the raw response string.

2	generateQRCode	String	Send a QR generation request to VietQR and return the raw response string.
3	checkPaymentStatus	String	Send a payment status or callback request to VietQR and return the raw response string.

Parameter:

getAccessToken(authorizationHeader : String) : String

- authorizationHeader: authorization header built from VietQR username and password, used to request an access token.

generateQRCode(accessToken : String, requestString : String) : String

- accessToken: valid VietQR access token used for authenticated API calls.
- requestString: formatted request payload containing bank and payment information for QR generation.

checkPaymentStatus(accessToken : String, requestString : String) : String

- accessToken: valid VietQR access token used for authenticated API calls.
- requestString: formatted request payload used to query or simulate payment status.

Method:

getAccessToken(authorizationHeader : String)

- Sends an HTTP request to the VietQR token API using the provided authorization header.
- Receives the raw response containing access token information.
- Returns the response string for further parsing.

generateQRCode(accessToken : String, requestString : String)

- Sends an HTTP request to the VietQR QR generation API using the access token and request payload.
- Receives the raw response containing generated QR information.
- Returns the response string for further parsing.

checkPaymentStatus(accessToken : String, requestString : String)

- Sends an HTTP request to the VietQR payment status or callback API using the given token and payload.
- Receives the raw response containing transaction status information.
- Returns the response string for further parsing.

Design for class “QRTestCallbackRequest”

QRTestCallbackRequest
- bankAccount : String - content : String - amount : long - bankCode : String - transType : String
~ QRTestCallbackRequest(bankAccount : String, content : String, amount : long, bankCode : String, transType : String) ~ buildRequestString() : String

Table 1. Attribute design

#	Name	Data type	Default value	Description
1	bankAccount	String		Bank account number used in the callback simulation request.
2	content	String		Payment content or transaction description associated with the order.
3	amount	long		Payment amount used in the callback request.
4	bankCode	String		Bank code of the receiving account.
5	transType	String		Type of transaction used in the callback simulation (e.g., incoming transfer).

Table 2. Operation design

#	Name	Return type	Description (purpose)
1	QRTestCallbackRequest	Constructor	Initialize the callback request object with payment and bank information.
2	buildRequestString	String	Build the request payload string for the VietQR test callback API.

Parameter:

QRTestCallbackRequest(bankAccount : String, content : String, amount : long, bankCode : String, transType : String)

- bankAccount: bank account number used in the callback request.
- content: payment description or order content.
- amount: transaction amount used for callback simulation.
- bankCode: bank code of the receiving account.
- transType: transaction type for the callback request.

Method:

QRTestCallbackRequest(bankAccount : String, content : String, amount : long, bankCode : String, transType : String)

- Initializes the callback request object with all required transaction information.
- Stores the provided values into the corresponding attributes.
- Prepares the object to be converted into a request payload string.

buildRequestString() : String

- Converts all stored attribute values into the format required by the VietQR callback API.
- Builds a formatted JSON / request string containing callback simulation data.
- Returns the final request string to be sent by the boundary layer.

Design for class “QRPaymentStatus”

QRPaymentStatus	
- message : String	
- status : String	
+ parseResponseString(response : String) : void	

Table 1. Attribute design

#	Name	Data type	Default value	Description
1	message	String		Message returned from the payment system describing the transaction result.
2	status	String		Current payment status (e.g., SUCCESS, FAILED, PENDING).

Table 2. Operation design

#	Name	Return type	Description (purpose)
1	parseResponseString	void	Initialize the callback request object with payment and bank information.

Parameter:

parseResponseString(response : String)

- response: Raw response string returned from the VietQR API containing payment status information.

Method:

parseResponseString(response : String): void

- Parses the raw response string received from the VietQR payment API.
- Extracts relevant information such as payment status and message from the response.
- Stores the extracted values into the corresponding attributes of the object for later use.